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Sustainability that backfires: the case of biogas in Emilia Romagna

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ABSTRACT

The development of a low carbon society does not always deliver sustainable outcomes at local level, as in the case of bioenergy. This paper analyses the impacts of biogas development in Emilia Romagna (Italy) from a social, economic and environmental perspective (triple bottom line). Since 2009, the number of biogas plants has exponentially increased here, following the introduction of the feed-in tariff. Social opposition has been rising since. Therefore, the goal of the study is to investigate the causal processes of biogas development in order to understand what endangered its triple bottom line sustainability. The study does so by employing qualitative system dynamics (i.e. causal loop diagram). Results show that sustainability is threatened by weak local synergies, plants management and equipment deficiencies and agriculture related issues. The analysis suggests that policy interventions should focus on biomass supply and outputs diversification, governance, and social participation.

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1. Introduction

In the last decade, the development of a low carbon and sustainable society has received enormous political and academic attention as it represents a desirable solution to climate change, energy security and economic growth (e.g. European Directive 2009/28/CE; European Commission, 2010a; European Commission, 2010b; Smith et al., 2005; Geels and Schot, 2007; Leach et al., 2010; Markard et al., 2012). Accordingly, the transition toward a low carbon society is supposed to lead toward a better future (e.g. renewable energy, electric cars) though in reality it may generate several negative outcomes e.g. social opposition, poor redistribution, higher energy bills, land pollution, bad smells, or competition with food production. The transition to bioenergy is a good example of this. Among other renewables, bioenergy has the greatest potential to be integrated in the rural economy, since it uses its endogenous resources such as land and agricultural products, human resources and local expertise (Bluemling et al., 2013; Del Rio and Burguillo, 2009; OECD, 2012; Raven and Gregersen, 2007; Raven and Verbong, 2004). However, evidence shows that what seems to be good at national level may become negative at local levels (Batel and Devine-Wright, 2014; Zoellner et al., 2008. In several cases the alleged benefits of bioenergy adoption (e.g. new job and business opportunities; capacity building and community empowerment, affordable energy and energy security) are not perceived, and the negative effects trigger local tensions and conflicts (OECD, 2012; Batel et al., 2013; Buchholz et al., 2009, 2007; Carrosio, 2013; Cavicchi et al., 2014; Domac et al., 2005; Geels and Raven, 2006; Mangoyama and Smith, 2011;

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Thornley et al., 2009; Upreti, 2004; Upreti and Van der Horst, 2004; Walter and Gutscher, 2010; Wüstenhagen et al., 2007). Yet, if such opposition emerges, as it has in many areas, it inevitably delays bioenergy adoption and increases the costs for bioenergy businesses, hindering the achievement of bioenergy goals. Bioenergy policies have rarely reckoned with the risks of linking such diverse systems – i.e. socio-economic and eco systems – without a serious account of potentially interrelated effects.

Therefore, the novelty of this study is twofold: on one hand, it questions the alleged benefits of bioenergy development and to investigate the causes for unsustainability. On the other hand, it considers social, environmental, economic activities and outcomes as mutually influenced via feedback relations. It thus draws on the case of biogas development (i.e. diffusion of biogas plants, biomass supply, electricity production) in Emilia Romagna (Italy) as an example of policy failure to support the adoption of a sustainable biogas system as per the triple bottom line (TBL) perspective of sustainability. The TBL considers environmental, social, economic activities and their outcomes as endogenously interrelated (e.g. Elkington, 1998; Bryden et al., 2008, 2011). The study uses qualitative system dynamics (i.e. causal loop diagram, CLD) to investigate these endogenous feedback relations. The CLD will map the causal processes of biogas development by using the information gathered through semi-structured interviews and public reports (i.e. regional database and statistical reports).

The paper is structured as follow: firstly, it presents the problem situation, research questions and objectives. Secondly, it introduces the theory, methodology and methods applied in the study. It then moves on into the results section, which includes the key facts, regulations and analysis of causal processes. The concluding discussion and policy recommendations highlight the key findings and potential policy implications.

1.1. Problem articulation: research questions, objective and structure

The current paper is based on a previous study of bioenergy policy in Emilia Romagna – and Norway – (Cavicchi et al., 2014; Cavicchi, 2013) that investigated the impact of policies and institutions on bioenergy and rural development in Emilia Romagna. Emilia Romagna hosts part of the largest plain in Italy (Padana Plain or Poí Valley) and quite a well-functioning regional agriculture. However since the introduction of the feed-in tariff at the beginning of 2009, the diffusion of biogas production has been causing several unforeseen problems (Cavicchi et al., 2014; Cavicchi, 2013; Carrosio, 2013). After initial high profit expectations, most of biogas operators are encountering economic difficulties. Additionally, people throughout the region have organized themselves in committees against biogas and perceive this new rural activity as extremely harmful for their health, the rural economy and the environment. The SWOT analysis in that study (Cavicchi et al., 2014) showed that obstacles to sustainable bioenergy development in the region were social opposition, the weak role of local governments, and weak links between government–businesses–research sector and negligible local benefits (Cavicchi et al., 2014; p. 362–363). The study looked at *bottom-up local participation* (p. 363)—although in the form of social opposition—as a potential trigger for policy and institutional change. Therefore, this paper asks: *what feedback relations influence the unfolding of biogas development in the case study? And which of them seem to foster or hamper its sustainable development (where sustainable means economic, social and environmental sustainability)?* By drawing on the triple bottom line concept, *"sustainability"* of biogas is understood as processes of its development that secure economic long-term feasibility to biogas producers while providing rural areas with social and environmental benefits, e.g. local GHG emissions reduction, new sources of income, local cooperation, etc. (see for instance OECD 2012; Bluemling et al., 2013; Del Rio and Burguillo, 2009; Mangoyama and Smith, 2011). In this sense, the paper particularly relies on the information gathered through semi-structured interviews with regional and local informants, public documents (e.g. regional policies – energy plans; resolutions, etc.) and relevant publications on Emilia Romagna and other cases (see for instance the OECD report 2012, Carrosio, 2013; Fabbri et al., 2013).

2. Theory and methodology

2.1. Theory

This study is predominantly an inductive 'ground-up' rather than a deductive 'theory-down' process, seeking to develop a contextual explanation of the causal processes of biogas development and sustainable outcomes in Emilia Romagna. It does not start off by hypothesizing a relationship between dependent and independent variables. Rather it aims at exploring the causes that link biogas development to unsustainable outcomes (dependent variable) in the case study. As such, the study's theoretical ground rests on the systems thinking tradition (Checkland, 1981; Senge 1990; Richardson, 1991; Forrester 1968; Sterman, 2000). Systems thinking is a broad and general approach to deal with a problem, whatever the subject area is. Two of the most influential scholars (Checkland, 1981; Senge, 1990) stress that "*systems thinking*" is a *"meta-discipline whose subject matter can be applied within virtually any other discipline"* (Checkland, 1981; p.5). Scholars within this tradition share a particular way of regarding the world, i.e. they assume that the world is made of 'wholes', which include social, biophysical, chemical etc. elements connected together.

There are different approaches within the systems thinking tradition. Literature commonly distinguishes between *hard-systems* and *soft systems* approaches. The former rest upon cybernetics and servomechanisms and employ mathematical modeling and simulations to seek the most efficient system's behavior solutions; the latter use a more interpretative–constructivist approach to enable an in-depth understanding of the organization's behavior (i.e. the system's behavior) to improve the way actors tackle critical organizational issues. However, there are significant overlaps

between the two, as shown by the intellectual and empirical work of several scholars (e.g. Checkland, 1981; Senge, 1990; Haraldsson, 2004; Haraldsson et al., 2006; Sterman, 2000; Lane, 1999, 2001). For instance, soft and hard approaches share the idea that '*systems*' (i.e. the set of causal processes underlying a certain problem situation) ensue from people's mental models and their interrelations. In other words, humans and social groups hold particular understandings and framings (i.e. mental models) of the world and problems around them that shape the trajectory of a system development (i.e. boundaries, rules, interactions, biophysical elements). The goal of both approaches is thus to investigate, case-by-case, the principles (or rules) that regulate the behavior of the system under investigation, namely how the social and biophysical elements relate to each other. System dynamics (seeForrester, 1968; Sterman, 2000, 2002; Ford, 2010; Lane, 1999, 2001, 2008) and Peter Senge's work (1990) go further and offer a more detailed theory of systems' behavior. The key building blocks rest on the concept of complexity, which arises from feedback, circular causality, and delays embedded in human actions. The concept of feedback simply means that an action is both cause and effect. Feedback loops are the product of the internal dynamics of a system, rather than ontological elements that exist in reality regardless of social perceptions and interactions (see for instance Richardson, 1991). Feedback loops then show how actions can reinforce or counteract (balance) each other; in other words, they display the underlying mechanisms (or structures) ensuing from people's mental models that regulate the system's behavior. Time-space delays (i.e. the effect of one event on another takes time to happen) are very common in human systems – for instance, we invest today to see a return in the future. Systems scholars assume that if delays are not appreciated, they will cause unexpected consequences that may overshoot the initial goals and disrupt the development of the system.

This study finds its theoretical ground somewhere in between soft and hard approaches. It fully embraces the view of the world as a complex whole made of interconnected social and physical elements. Particularly, the expression *"biogas system"* refers to the complex interrelation of causal processes that include social, economic and environmental, as well as material variables (e.g. social conflicts, profit, investments, GHG emissions and local warming, farmland). These processes are thought to be the result of regulations, actors' decisions and agency, knowledge interchanges between them and material processes (e.g. anaerobic digestion, biogas combustion, electricity transmission, etc.). Therefore, the study adopts the assumptions of system dynamics about complexity, feedback causality, and time-space delays. However, it builds on the subjective-interpretative stance to enable organizational learning through causal mapping. This theoretical approach may contribute to the socio-technical transitions (e.g. Geels, 2002; Geels and Schot, 2007; Markard et al., 2012; Coenen et al., 2012) with a stronger conceptualization and operationalization of feedback relations between the factors that influence systemic transitions, i.e. institutions, actors, networks, market forces, legitimacy issues, biophysical, technological and infrastructural conditions. This could enrich the investigation of how new socio-technical systems emerge, but also enable a policy-making based on a more thorough consideration of complexity and consequences of feedback. The conclusion paragraph will elaborate this point further based on the empirical findings.

2.2. Methodology and methods

In line with the theoretical background, this study employs qualitative system analysis to investigate the unfolding processes of biogas development and its outcomes in Emilia Romagna (Haraldsson 2004; Lane 1999, 2001, 2008; Luna-Reyes and Andersen, 2003; Meadow et al., 1972; Senge, 1990). Qualitative systems analysis is believed to enable the in-depth investigation of causal processes as ensuing from different actors' visions and decisions. Other scholars have used causal loop diagrams to explore and communicate the feedback processes underlying social dynamics (see for instance the work of Haraldsson, 2004; Maani, 2002; Maani and Cavana, 2007; Spector et al., 2001; Cavana and Mares, 2004; Haraldsson et al., 2006). These studies stress that although simulations provide significant and rigorous insights to qualitative mapping, simulation results are difficult to illustrate and communicate clearly to different types of audience. Despite their weaknesses (e.g. flows and time delays cannot be properly included in the model), causal loop diagrams help the practitioner understand how different stakeholders perceive a problem and their different mental models (i.e. visions) interact to shape the system's processes and outcomes. It is also acknowledged that CLDs are an oversimplification of reality but that, as such, may enable a clearer understanding of the problem. Thus, CLDs assist and enhance issue structuring, organizational learning and problem solving (see for instance Checkland, 1981; Senge, 1990; Wolstenholme, 1999). In this sense, CLD is considered to be a good tool to best analyze and illustrate the structure of biogas development in Emilia Romagna as the result of feedback relations between institutional arrangements (i.e. policies, regulations, governance), stakeholders' perceptions and actions, and biophysical elements (e.g. farmland, irrigation ditches, etc.). This will also provide the basis to draw some general policy recommendations based on in-depth learning and knowledge building.

2.2.1. Methods and data sources

As to assure the degree of validity and reliability of this qualitative study of biogas development in Emilia Romagna, this section presents the methods, iterative and reflexive process that led to the empirical findings (Noble and Smith, 2016; Shenton, 2004; Golafshani, 2003; Silverman, 2001; Yin, 1994).

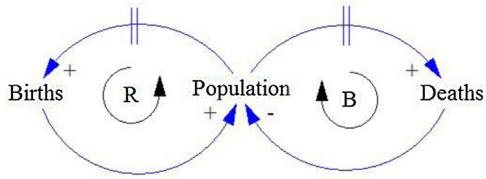

Fig. 1. Example of Causal Loop Diagram.[1]

Source: Sterman, 2000

Table 1
Key informants interviewed.

Categories	**Number**
Bioenergy businesses	14
Farmers' unions	5
Biogas and sectorial organizations (CIB and CRPA)	3
Local and regional governmental actors (i.e. local and regional decision makers, environmental agency)	8
Experts (i.e. professors and researchers)	5
Incubators and innovation centres	2
Local committees	4

The study employs causal loop diagram that is drawn with VENSIM software. Fig. 1 shows a basic causal loop diagram structure.

Causal loop diagrams indicate feedback relations between variables and are usually employed for descriptive purposes. Feedback loops can be reinforcing (R) and produce growth, or balancing (B) and stabilizing loops. Feedback loop is reinforcing when variables are positively related and their value goes in the same direction (+); feedback loop is balancing when variables are inversely related and the value goes in the opposite direction (−).[2] Arrows link variables and indicate the causal influence between them (+, −). Once the arrows and loop polarity have been assigned, they should not need to be changed; wherever the loop reading may be started. Delays are also very important to place, as they indicate the time expansion of consequences of actions. Delays come in the relation between a flow and a stock, i.e. the example above tells that it takes time before a growing population can affect, for instance, the births rate. This delay usually generates oscillations and potential unexpected outcomes, between birth and death rates (e.g. a sudden famine could kill many people thus decreasing birth rate).

In the study, the causal loop diagram is the output of data aggregation that includes information from semi-structured interviews (Table 1 below), public documents related to the case study (in particular Moratorium, 2011; NREAP, 2011; PER, 2007, 2011; PSRR 2007; Regional Act, 26/ 2004), reports and relevant publications on the case study (OECD, 2012; Carrosio, 2013; Cavicchi et al., 2014; CRPA, 2008; Fabbri et al., 2013). Semi-structured interviews are the main data collection method and were conducted between April 2014 and March 2015. The selection of informants was based on the author's previous study on bioenergy policy in Emilia Romagna (Cavicchi, 2013; Cavicchi et al., 2014) and integrated with new informants selected through the 'snow-ball' method – which relies on suggestions and cooperation of the previous interviewees. Random sampling "[. . .] *may negate charges of researcher bias in the selection of participants. Random sampling also helps to ensure that any 'unknown influences' are distributed evenly within the sample.*" (Shenton, 2004; p.65). However, since informants' suggestions may also be biased, the selection was integrated and crosschecked with existent relevant studies (e.g. Carrosio, 2013; Fabbri et al., 2013) and regional database of biogas firms.[3] Informants were asked specific questions on their main activities in relation to the problem situation (e.g. biogas production, plants inspection, research, advise, etc.) and to give their perspective on advantages and disadvantages of biogas production, obstacles to sustainable biogas production and, on the other hand, their suggestions to make it more sustainable.

This raw data was processed into the causal loop diagram through an iterative and reflexive process (see Alvesson and Sköldberg, 2009; Shenton, 2004). This means that both data collection and the construction of the causal loop were repeatedly checked and updated until the final version of the CLD was exhaustive and did not raise other doubts or showed inconsistencies. The construction of the CLD is based on the triple bottom line. Hence, the data collected was coded into economic, social and environmental causal processes based on the respondents' narratives, public documents and reports. This codification was complemented with reference to the early adopter of the triple bottom line approach (Elkington, 1998) and a relevant system dynamics study that applied the TBL to investigate the sustainability of multi-functional agriculture (Bryden et al., 2011).

[1] Fig. 1 above must thus be read as follow: in the absence of external factors, by time population leads to more births that feeds back by increasing the initial population (Reinforcing loop, +, +). However, more population also leads to more deaths and so to a reduction in population (Balancing, +, −).

[2] An example of the first can simply be the more a company sales (+), the more revenues they get (+). An example of the second is if the number of predators increases (+), the number of preys decreases (−).

[3] https://www.google.com/fusiontables/DataSource?docid= 11WLPSgNkbLATOha_CMDhNOWEzRGfml-RitPqWbP0#rows:id=1

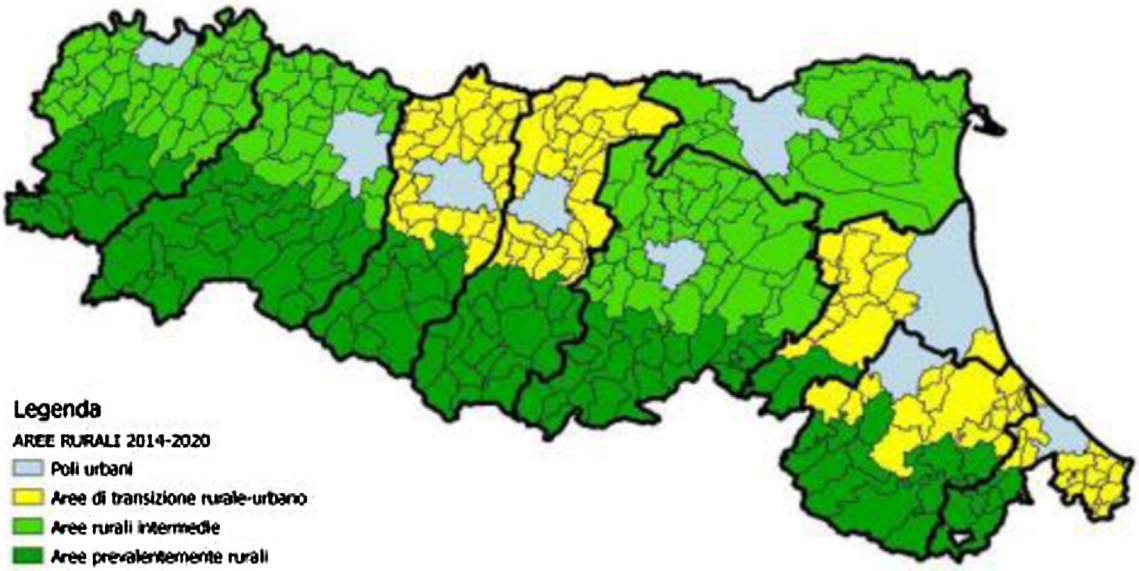

Fig. 2. Regional Rural Areas classification.[4]

Therefore, the first draft of the CLD was drawn based on the information gathered in the author's previous study (Cavicchi et al., 2014; Cavicchi, 2013). For instance, this included basic information about the feed-in tariff scheme and related investments in biogas, local opposition, bad smells, traffic, the relevant actors and main public documents. However, the information provided by different actors at the beginning of the new interviewing round showed that some causal relations were inconsistent or not complete (e.g. rise in farmland rent price; percolation into irrigation ditches; local warming; local opposition). This required an in-depth revision of the CLD via new interviews and updates to clarify doubts and contradictions. At the same time, this process of revision was crosschecked via regional statistics, public documents and reports that could shed light on the most questionable issues (for instance the price of farmland rent and increase in energy crops production). Further, previous literature on bioenergy and sustainable rural development (in particular OECD report, 2012 Wüstenhagen et al., 2007; Mangoyama and Smith, 2011; Domac et al., 2005; Upreti, 2004; Upreti and Van der Horst, 2004; Elkington, 1998; Bryden et al., 2013; Rogers and Ryan, 2010; Stanzel, 2010) provided useful material to scrutinize the informants' narratives and the researcher's interpretations with other cases. Therefore, the CLD's variables represent the key points emerged in the interview material (i.e. feed-in tariff, farmland rent price, local warming, traffic, etc.) as crosschecked via the iterative process and multiple data sources previously described. This iterative process can be used to monitor the reliability and validity of the causal loops (see for explanation Haraldsson, 2004). However, this also required discussions with colleagues, in particular my supervisors and an expert of qualitative system dynamics in Sweden. Moreover, the study was presented at two conferences, one in Bergamo (June 2015) and one in Florence (September 2015).

3. Results

3.1. The case study context

Emilia Romagna is a Northeastern Italian region which territory extends for 22,452,78 km^2 and accounts for 4,446,354 citizens that inhabit 340 municipalities (198 inh/km^2 on average) (ISTAT 2014). The regional territory is circa 48% flat and hosts the biggest and most fertile Plain in Italy (southern part of the Po' Valley). The rest of the land is split in hilly areas (27.1%) and mountain areas (25.1%). Most of the regional population (64%) and land (90%) is defined as rural. Of this rural population the 76.7% inhabits intermediate rural areas (mostly located around urban centres); whereas rural areas classified as principally rural and with development problems account for 36.7% of the regional land and are mostly mountainous (see Fig. 2)

Emilia-Romagna is one of the most industrialized regions in Italy and contributes for about 4.3% of national exports, especially in vehicles (18.3%). Manufacturing and services are the leading regional economic sectors, accounting for 29.57% and 54.95% of the the regional value added in 2013 respectively. Agriculture is still relevant in the regional economy accounting for the 2.77% of the regional value added in 2013 and showing a steadily increasing trend from 2011 (2.49%). In terms of production, since the adoption of the 2004CAP reform (lower support for arable crops and sugar beet) data shows a

[4] http://www.reterurale.it/flex/cm/pages/ServeBLOB.php/L/IT/IDPagina/11182

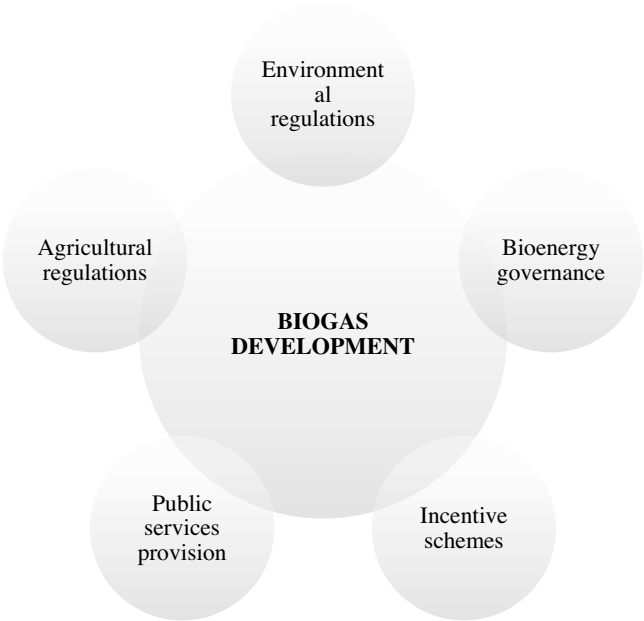

Fig. 3. Regulations impacting on biogas development.

stable increase in food crops production (e.g. wheat) over industrial crops (i.e. sugar beet). Statistics also show an acquisition tendency of smaller farms by larger farms and an increasing share of rented farmland (ISTAT, 2010). Regional farmers are organized into three farmers' unions, i.e. Coldiretti, Confederazione Italiana Agricoltori and Confagricultura. The latter represents big-size and industrial-related farms (food industry and ex-sugar industry); whereas the two former represent mostly small farmers. Moreover, the regional Centre for Animal Production (CRPA)[5] is the main actor engaged in research on farm/agriculture-based biogas. The center is also a knowledge incubator and learning platform where farmers can meet and exchange experiences, for instance in terms of production processes, and biomass quality.

The regional energy system is mostly based on natural gas and oil as for both electricity and heat generation (same figure at the national level). In Italy, natural gas and oil are imported from abroad (Russia, Algeria and Norway) for about 65% of total Mtoe production (AEEG 2015b).[6] National figures show that natural gas and oil account for 68.1% of energy consumption while renewable energy accounts for 18% indicating an increasing trend compared to 2012 (15.1%) (AEEG, 2015a). Available regional figures show that in 2012, 13.1% of the total natural gas distributed per region (in m^3) went to Emilia-Romagna, second only after Lombardia (26.6%) (AEEG, 2013e). The Italian energy market was initially based on two national monopolies in natural gas and electricity, ENI and ENEL, respectively until the liberalization of the 1990s. Currently, the energy market shows the features of an oligopoly (especially in electricity production and import of natural gas). Nationally, ENI is the major natural gas importer (circa 50%) followed by Edison and ENEL (AEEG, 2015c); whereas ENEL is the major electricity producer (25.4%) followed by ENI and Edison (AEEG, 2013b). ENI and ENEL are also the major renewable energy producers (AEEG, 2013d). The natural gas market is rather decentralized in terms of distribution (236 distributors nationally). The main natural gas distributor in the country is SNAM rete gas (23.1%), whereas in the region there are other two major distributors, i.e. HERA (6.5% of m^3) and IREN (5.9% of m^3) (AEEG, 2013c). HERA is also one of the biggest national producers of bioenergy (5.3%) (AEEG, 2013a,d).

3.1.1. Regulatory context

Fig. 3 shows the regulations that influence the trajectory of biogas development. These are environmental and agricultural regulations and norms; incentive schemes; regulations on public service provision and bioenergy governance.

The interest of regional stakeholders (e.g. farmers, manufacturers and Centre for Animal Research – CRPA) in biogas dates back to the 1970s. The oil crisis and the need to process manure in livestock farms were the drivers of this initial effort to develop biogas technology internally. However, it was only at the outset of the 2000s, following the European policy developments (i.e. CAP reform 2004; European Directive, 2001/77/EC on renewable energy and European Directive, 2003/30/CE on biofuels) and national transpositions (i.e. DL, 387/2003) that the region laid the first consistent regulatory basis for biogas development. On one Hand the adoption of the 2004CAP reform imposed a restructuration of the sugar industry, which was one of the main regional economic sectors. As a consequence, most of the sugar factories and sugar beets

[5] http://www.crpa.it/nqcontent.cfm?a_id=1109
[6] National extraction has been in constant decline since at least 1997 and nowadays accounts only for 7% of the total amount.

cultivation closed down and farmers opted to use part of the arable land to cultivate food crops such as wheat (CRPA, 2008:9). On the other Hand, the DL, 387/2003 (updated by Dlgs, 28/2011) adopting the EU Directive 2001/77/CE and 2003/30/CE, for the first time stated that renewable energy facilities are of public interest and thus urgent and not deferrable. These national evolutions led to the Regional Resolution 26/2004 that still regulates the energy sector and renewable energy development in Emilia Romagna. The Resolution set the regional priorities to be energy efficiency and renewable energy production, but most importantly it shaped the public governance of regional energy planning (Art. 2, 3, 4). Notably, regional and provincial level were assigned the key responsibilities on energy planning (e.g. concessions), whereas municipalities were left with residual duties on energy efficiency and district heating matters (e.g. support the development of the infrastructure). In greater detail, the reason for the minor role of municipalities in energy matters is rooted in the 1990s, when public services such as waste and water management, natural gas and energy supply, were privatized. As a consequence, at the beginning of the 2000s the municipal public companies in Emilia Romagna merged and created two private multi-utilities, HERA and IREN, which provide public services.

In 2007 the regional energy plan—PER—(one of the first in Italy) indicated biogas and solar energy as the main regional alternatives to natural gas. On this account, the national and regional rural development plans 2007–2013 (following the EU Regulation on Rural Development 1698/2005) strengthened the link 'bioenergy-rural development' by stressing its potential benefits for rural employment and energy diversification (PER, 2007, p. 412). The key driver for bioenergy development was introduced at the beginning of 2009, namely the all-inclusive feed-in tariff scheme (AI-FIT). The AI-FIT is a national scheme applicable to renewable energy plants (excluding solar ones), which have a nominal real power of less than 1 MW (200 MW for on-shore wind plants). The tariff is granted over a period of 15 years for all plants commissioned by 31 December 2012, during which its rate remains fixed and based on the amount of electricity fed into the grid. The incentive cannot be combined with others, but it makes an exception to farm-based plants that can combine it with other national, regional or local incentives for about up to 40% of the total investment cost. In addition to the AI-FIT, a tax exemption also applied to farm-based bioenergy production between 2006 and 2014.[7] In order to be granted the tax exemption, biogas producers must register as farmers in the municipal register. To be registered as farmers, they have to prove that at least 51% of the biomass fed in the plant is from the biogas firm's fields.[8]

The beginning of the 2010s saw some further specifications in the national regulations that followed the transposition of the RES Directive 2009/28/CE. For instance, the National Guidelines for Renewable Energy Production (Decree September 10th, 2010) established that only regions and autonomous provinces can limit the installation of specific typology of renewable energy plants and take part of the concession process (i.e. provincial department of environment, regional and provincial environmental agency—ARPA—provincial health agency—USL) (Art. 1.2). Municipalities were again marginalized by national regulations. In 2011 Emilia Romagna introduced both new and stricter environmental regulations on the edge of the arising social opposition and adopted a new regional energy plan (2011–2013). The latter set the regional 2020 renewable energy targets—adapting the national target (17%)—to regional conditions (i.e. available resources and knowledge). Photovoltaic is expected to contribute to electricity generation to the highest extent, followed by biomass, whereas biomass should take the highest share in the heating sector (PER, 2011; p. 110) . Yet, at the end of 2012 the national government modified the AI-FIT scheme that now differentiates the tariff amount per biomass type. It thus allocates the highest tariff amount to agriculture byproducts rather than to energy crops. In the same year, the government also established the regional burden sharing, i.e. the regional RES quota. Emilia Romagna should increase its RES up to 8.9% by 2020 compared to level of 2005. However, the target is not mandatory. Finally, between 2014 and 2015, the national authorities gave dispositions (i.e. quality and technical criteria; incentive scheme) for the production and transmission of biomethane from biogas in the natural gas grid (DM 2013; Resolution, 46/ 2015).

3.2. Processes and outcomes of biogas development

This section examines the processes and sustainable outcomes of biogas development in Emilia Romagna by using a causal loop diagram (Fig. 4). Blue arrows and black variables indicate the economic-production processes and outcomes; the green ones stand for environmental processes and outcomes; the red ones indicate social and political processes and outcomes; lastly the pink ones are about technology-equipment factors. The use of different colors is thought to help reading through the diagram and make the interrelations of different processes (i.e. economic, political, social, environmental and technical) and feedback more explicit and clear.

The closure of the sugar sector in the aftermath of the 2004 EU CAP reform and the adoption of the 2020 targets in 2009 pressured the national and Emilia Romagna's government to intervene (see for instance CRPA 2008; interviews with CRPA). Informants within farmers' unions and biogas firms say that the introduction of the "all-inclusive feed-in tariff scheme" (AI-FIT) and the tax exemption on farm-based bioenergy production persuaded farmers and other industrialists (i.e. from the sugar industry) to invest in biogas. According to the regional database, between 2010 and 2012 the number of biogas

[7] http://www.agenziaentrate.gov.it/wps/file/nsilib/nsi/documentazione/provvedimenti + circolari + e + risoluzioni/circolari/archivio + circolari/circolari + 2009/luglio + 2009/circolare + 32 + del + 06 + 07 + 2009/circ + +n + +32E + del + 6 + luglio + 2009.pdf; http://www.gazzettaufficiale.it/eli/id/2014/04/24/14G00079/sg

[8] http://www.crpa.it/nqcontent.cfm?a_id=7233&answer5; http://www.gse.it/it/salastampa/GSE_Documenti/Guida%20Fonti%20Rinnovabili.pdf p.19

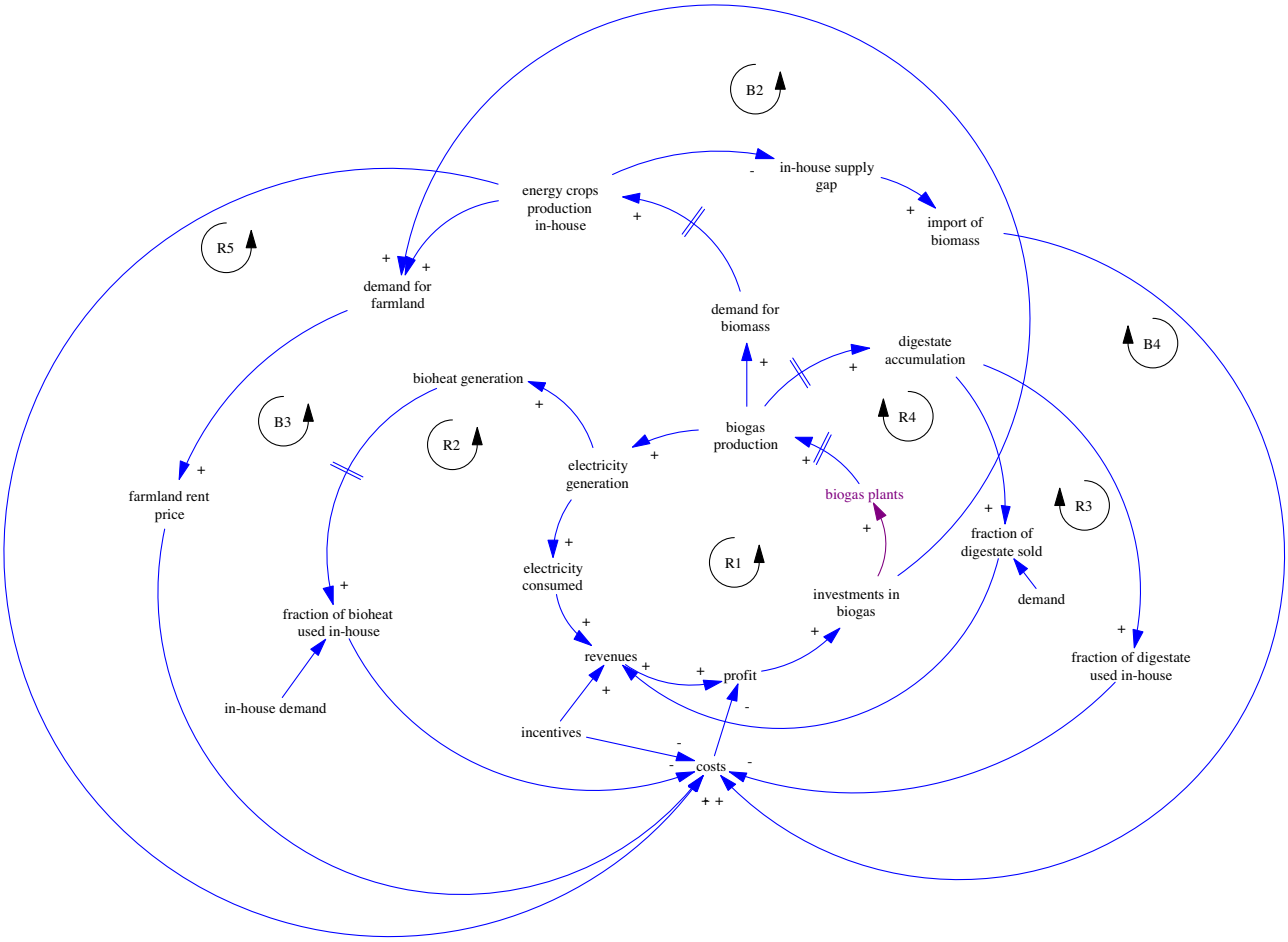

Fig. 4. CLD Economic causal loops. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

plants increased exponentially, going from approximately 29 plants to 172.[9] As Fig. 4 shows, the byproducts of biogas production in Emilia Romagna are electricity, digestate and heat. The feed-in tariff scheme remunerates biogas producers for the electricity that they generate up to the threshold of 999 kWh (R1).[10] The exceeding part can be used in-house or sold in the market. Biogas producers normally use digestate in-house as an organic fertilizer but they may also sell it to other farmers (interviews with farmers' unions and biogas firms). In each case, biogas producers make a profit by either reducing the costs (R4) or producing revenues (R3). Finally, biogas firms consume part of the heat in-house thus reducing the cost of energy supply (R2). However most of the heat remains unused and thus is dispersed in the air (B1).

As a consequence of adopting biogas plants, investors were often forced to rent new farmland or dedicate most of their own to the installation of the plant and related storing facilities (interviews with biogas firms and farmers' unions).[11] The interviewees agree on that this led to an increase in farmland rent price impacting on the overall costs and profit of both biogas producers and surrounding farmers (B2, B3). Farmers' unions suggest that the growing production of energy crops (e.g. sorghum and triticale) also affected the increase in farmland rent price. These assertions seem to be supported by recent statistics of the Italian Agricultural Economics Institute (INEA, 2013 and historical database).[12] and Emilia Romagna Region[13] (2015 and historical database)–in addition to specialized newspaper articles.[14] The former source suggests a rent price increase between 2008 and 2012, in line with the regional database that shows an increase in farmland rent and a

[9] ARPA's https://www.google.com/fusiontables/DataSource?docid=11WLPSgNkbLATOha_CMDhNOWEzRGfml-RitPqWbP0#rows:id=1.
[10] http://www.gse.it/it/Qualifiche%20e%20certificati/Tariffa%20omnicomprensiva/Pages/default.aspx
[11] Research shows that ca. 400–500 ha are needed to run a 999 KwH plant, of this ca. 245 ha should be used to produce the necessary amount of energy crops, ca. 12.245tons/year. Source: CRPA, Líinformatore agrario 2011
[12] http://www.inea.it:8080/mercato-fondiario/esempi (especially on land rent prices)
[13] http://territorio.regione.emilia-romagna.it/osservatorio/notizie/valori-agricoli-medi-2015 (especially on land value in euro per hectare)
[14] http://agronotizie.imagelinenetwork.com/agricoltura-economia-politica/2011/07/21/mercato-fondiario-e-affitti-l-analisi-di-inea/13735, http://www.conipiediperterra.com/inea-la-crisi-pesa-sul-mercato-fondiario-meglio-gli-affitti-0725.html.

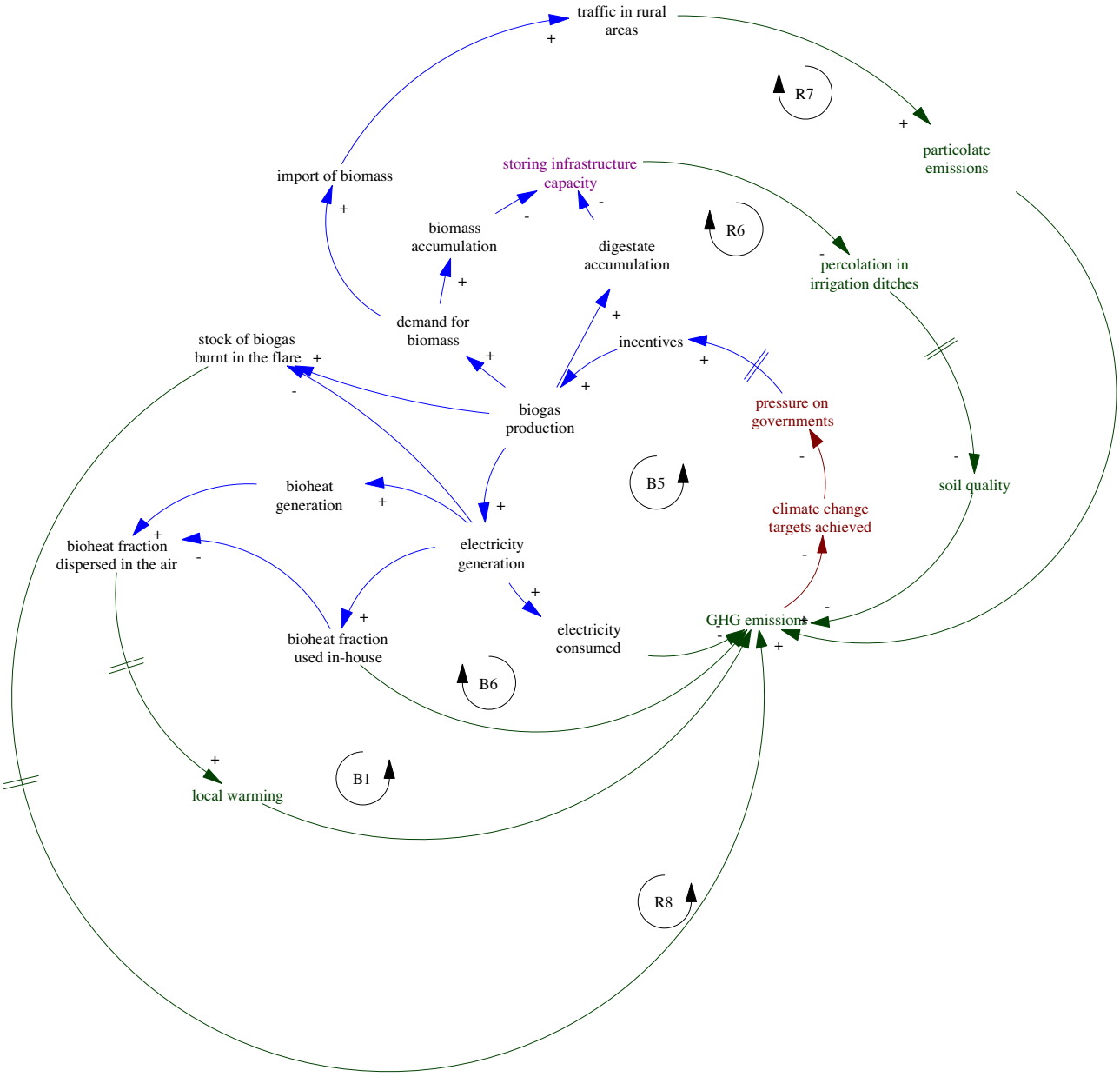

Fig. 5. CLD Environmental causal loops.

decrease in farmland value[15] Most generally, biogas producers decided to cultivate energy crops in-house to reduce the costs related to biomass supply (R5), but also to meet the regulatory requirement (51% of the biomass must be from the firm's fields) that allows for the tax exemption on farm-based biogas. However, informants lament that the unstable weather of the last few years frequently damaged the harvesting forcing them to buy biomass from other suppliers, thus increasing the costs for the firm (B4) (interviews with biogas producers and farmers' unions).

The environmental effects of biogas development are both positive and negative (Fig. 5). The transmission of electricity in the grid, the consumption of 'bioheat' in-house and the use of digestate contribute to GHG emissions reduction, thus releasing pressure on the regional government (B5). However, negative outcomes result from biomass and digestate accumulation (R6), traffic (R7) and over-production of biogas (R8, B1). First, many biogas firms lack of proper storage arrangements (i.e. facilities, exact amount of time), thus for instance, the liquid that results from biomass decomposition (leachate) does in

[15] http://territorio.regione.emilia-romagna.it/osservatorio/notizie/valori-agricoli-medi-2015 and http://statistica.regione.emilia-romagna.it/servizi-online/censimenti/6b0-censimento-dellagricoltura-2010

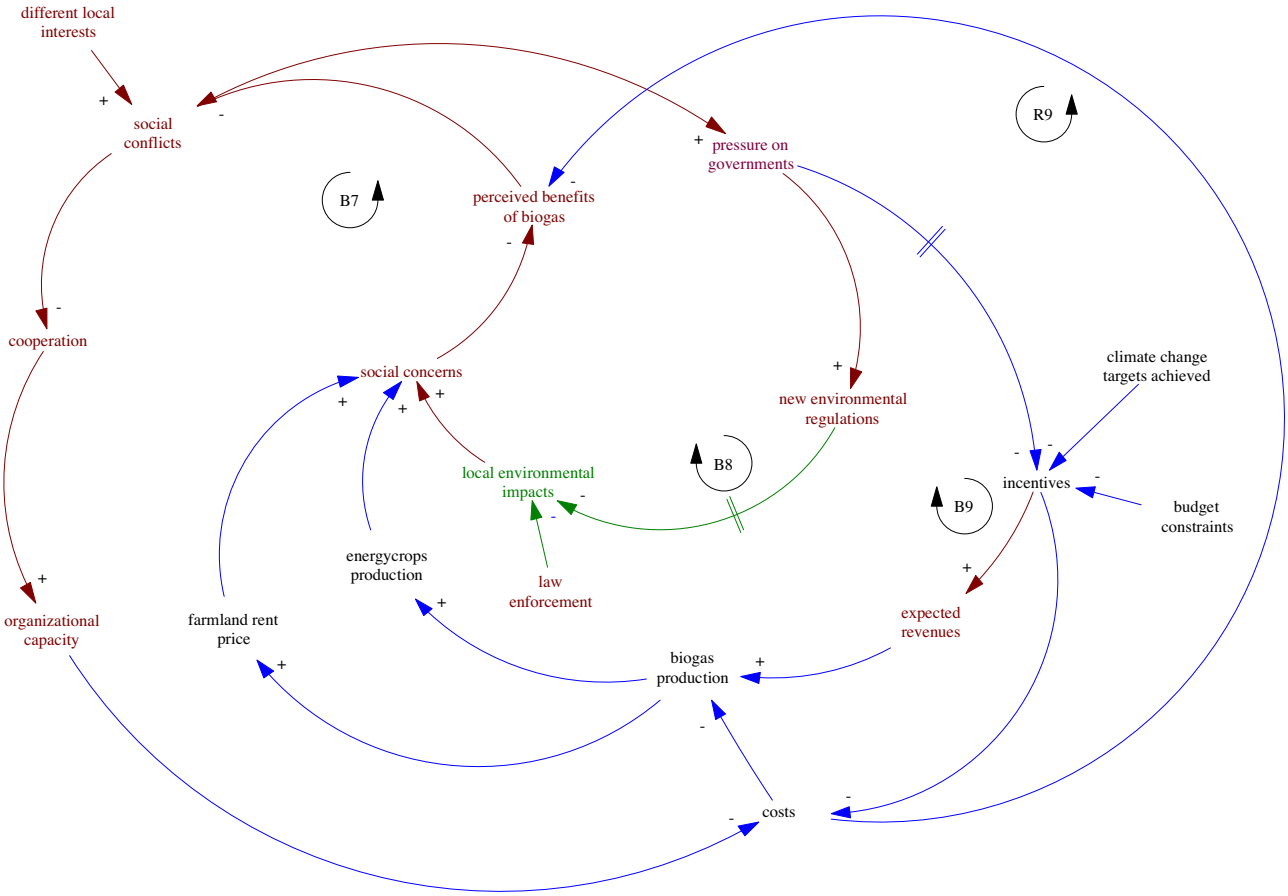

Fig. 6. Social causal loops.

many cases flow into irrigation ditches (interviews with local committees and ARPA). Leachate affects soil and air quality (including bad smell), thus eventually resulting in more GHG emissions (R6). Second, since the amount of electricity fed in the grid is conditional to the feed-in tariff (R1), it may happen that the exceeding part of biogas is burnt in the flare thus contributing to GHG emissions increase (R8). Third, local inhabitants (interviews with Legambiente and local committees) claim that traffic has increased within biogas locations as a consequence of transport of biomass from other areas (R7). If this holds true there may be an increase of particulate and GHG emissions. Finally, the heat dispersed in the air is likely to increase local warming and GHG emissions connected to this (B1) (interviews with Legambiente and ARPA). Biogas producers claim that the plants are located too far from other rural buildings for an efficient transmission (i.e. reduced loss), whereas the absence of an incentive makes it unaffordable.

Since 2009, the economic and environmental effects of biogas production have raised social concerns (e.g. Moratorium, 2011; Legambiente, 2012). In some cases (e.g. Province of Ferrara and Bologna), local inhabitants have formed local committees to fight against the uncontrolled diffusion of biogas plants (interviews with Legambiente and local committees; Moratorium, 2011). Fig. 6 illustrates how these actions have strongly influenced people's perception of the benefits of biogas and its social acceptance. Thus, social conflicts have increased, especially between farmers, local inhabitants, municipal authorities and biogas producers (B7, Fig. 6). For instance, local inhabitants often claim that they were not informed or involved in participatory processes or discussions about new biogas projects (interviews with local committees and legambiente). Yet, local synergies are almost absent, especially if one looks at at the low number of biogas cooperatives compared to that of individual or smallfirms (only 8–10 cooperatives out of 172 plants, ARPA Emilia Romagna, 2013). This seems quite at odds with the traditional system of cooperatives and unions that characterize Emilia Romagna (see for instance Brusco, 1982; Bianchi and Giordani, 1993). Although it is difficult to establish a causal relationship, informants suggest that this lack of cooperation impacts negatively on the organizational capacity of the biogas sector thus also increasing the costs for biogas producers (B7). In 2011, the social pressure led the regional government to regulate the procedures for digestate and biomass storage and plants location (Delibera, 51/2011; Delibera, 969/2011; Delibera, 1495/2011) (interview with Regional Department of Economic Development). Loops B8 show that these interventions could potentially reduce GHG emissions thus releasing political pressure. Meanwhile, the national government modified the AI-FIT at the end of 2012 (B9). Interviews suggest that it was the interrelation of the social pressure represented by regional stakeholders in the parliament's

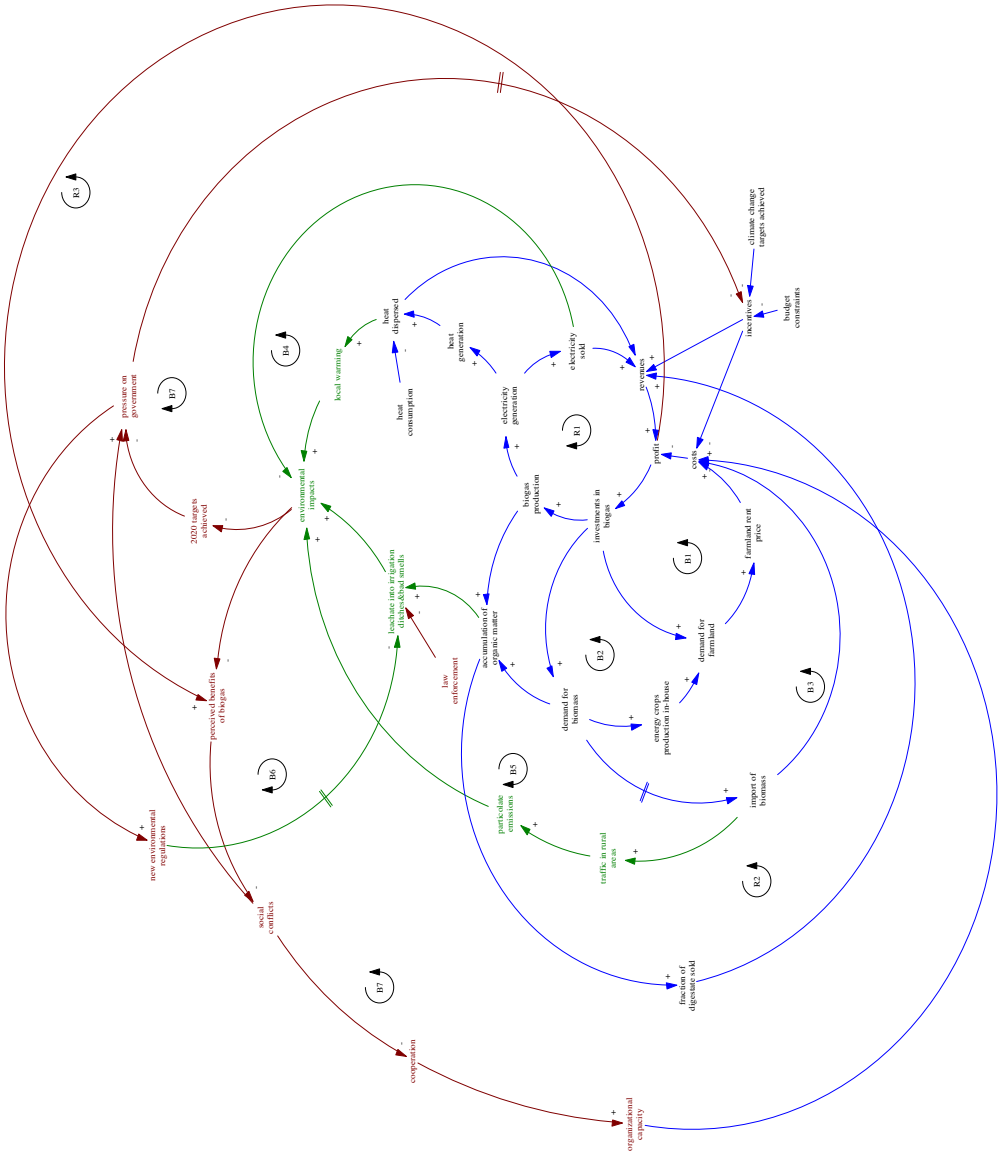

Fig. 7. Complete CLD of biogas development.

commissions (e.g. farmers unions and regional governments), budget constraints and achievement of the EU 2020 goals (see EU tracking road map, p. 58) to trigger the decision on the AI-FIT scheme. Interviews with ARPA and farmers' unions, and the evidence gathered in ARPA's database[16] would indicate a slow down in biogas investments that followed the new FIT scheme.

Lastly, Fig. 7 shows the core feedback relations between economic, environmental and social processes.

4. Discussion

The analysis has shown that there is a case for unexpected consequences and 'overshooting' in Emilia Romagna. The attempt to support farmers and tackle climate change issues with short-term policy solutions has backfired and worsened the initial conditions (see Senge, 2011). The model here below sums up these key causal interrelations (N.B. loops number do not correspond with previous diagrams since this one below is not as detailed as the others). Notably, the policy goal was to revitalize the northeastern agriculture sector in Italy via biogas production. This should have fostered the diversification

[16] https://www.google.com/fusiontables/DataSource?docid=11WLPSgNkbLATOha_CMDhNOWEzRGfml-RitPqWbP0#rows:id=1.

of agriculture primary productions and the integration of the farmers' income, while also contributing to the achievement of the EU 2020 targets. Although the region has strongly contributed to reach the 2020 national renewable energy share, the accomplishment of climate change mitigation and rural development appears more controversial. The key hypothesis is that the causes of unsustainable triple bottom line outcomes are, in particular, the land-related and biomass supply loops (i.e. rent price increase; energy crops production and import of biomass; biomass storage) and perceived benefits of bioenergy loop (i.e. decrease of social acceptance). The former triggered further processes such as local warming, particulate emissions and leachate percolation in irrigation ditches, and social conflicts (i.e. opposition from local inhabitants, conflicts between small and big farmers) that fed back into pressure on regional and national governments. Therefore, while the production of biogas-based electricity contributes the EU 2020 targets, it does also produce local environmental and social consequences. Finally, the potentials of biogas are not fully harnessed without its use in transports and the consumption of heat. Governments responded with new environmental regulations and revisions of the incentive schemes. It is too soon to assess whether these interventions have benefited or not the sustainability of biogas development. However, there have been some visible changes such as ARPA's inspections have become more thorough and strict (interviews with ARPA) while the new incentives have slowed down investments in biogas.

Based on these empirical findings, some questions arise such as: 'is it possible to manage the undesirable consequences? Are there alternative "fixes" in which the unexpected and undesirable consequences can be less devastating or are the interventions needed at all?' Senge (2011) argues that in these cases it is wise trying to reframe and address the root problem rather than only the symptoms. *"Every fix that backfires is driven by an implicit target in the balancing loop. So make it explicit. What's the problem you are really trying to fix?"* (p.129). The answer to these questions is that at the beginning the problem was the crisis of the agriculture sector in Emilia Romagna and Northeast but now is really social opposition and more in general the social conflicts that arose in the aftermath of the diffusion of biogas plants.

5. Conclusion and policy recommendations

The paper argued against the assumption that bioenergy development does in any case deliver sustainable outcomes embodied in political strategies. Sustainability was here understood as triple bottom line, i.e. as the result of interrelated economic, social and environmental processes. The study thus drew on a regional case study (Emilia-Romagna, Italy) of biogas development and explored the causes for its unsustainable outcomes, i.e. strong social opposition to biogas, environmental and economic drawbacks. The systems thinking theory presented in the paper assumes that unexpected outcomes ensue from circular causal processes that take time to fully unfold. These causal processes are regulated by feedback that may either reinforce or balance the initial system's conditions. Therefore, due to time-space delays, systems may overshoot the desired goal(s). The exploration of causal processes of biogas development contributes to the understanding of the interrelated processes that lead to [un]sustainable outcomes; as such, it consequently broadens the field of sustainability transitions.

Firstly, this approach brings the *feedback* concept at the center stage of the so called 'sustainability transitions' (ST). Although in 'sustainability transitions' the single factors affecting the establishment of new socio-technical regimes may be quite well-known (i.e. institutions, legitimacy, market dynamics, biophysical limits), the *feedback relations*, i.e. how they mutually influence each other, are not as well disclosed as they could be. In particular, the feedbacks between environmental-social-economic processes needed deeper exploration. Therefore this study can contribute to ST by integrating the economic and social processes with the biophysical, infrastructural and environmental processes to explore how these endogenous dynamics affect the expected sustainable outcomes of the transition and the establishment of an emergent system. Moreover, this approach includes in the model the feedbacks between 'landscape' conditions (in this case EU 2020 targets) and the inner levels of niche and socio-technical regime.

Secondly and in relation to the questions raised in the discussion, there are some important policy considerations. The incentive scheme was a crucial problem because it boosted profit expectations in a context where there were conflicting interests and little appetite for cooperation. Politicians thought that an economic incentive could accelerate the achievement of policy goals and did not appreciate the consequences that this one-way strategy could entail. One crucial recommendation in this sense is that policy portfolio should be diverse and simultaneously address different issues. As Fig. 7 shows, policy could target especially biomass supply, cooperation-synergies and output diversification of biogas production. Outputs diversification could generate other sources of income for biogas owners and thus increase their revenues (see loop R1). This could also mitigate the expected drop in the profit of biogas firms at the end of the feed-in tariff scheme (R1). However, this should be coupled with actions that enhance cooperation and synergies at local and regional level and the demand for other biogas plants products, such as bioheat and biomethane (see loop B7). For instance, the main reason for not supplying heat is that it lacks users, as plants are located too far from residential and industrial buildings and private households are not interested in changing heating source. Local and regional governments could support projects that foster knowledge-transfer, and platform building to inform private households and local biogas producers of the benefits of cooperation on infrastructural investments. Another way to stimulate sustainability could be to set ambitious climate change goals, as for instance the mandatory goal to reduce fossil fuels quota in public buildings and transports. This should lead to use alternative energy solutions such as biogas, at least in the areas where there are several biogas plants.

The same approach may apply to biomass supply. Alongside economic instruments, interventions that foster local synergies (e.g. between farmers, biogas producers, food industries and waste companies) could result in shared costs and thus

reduce the burden of biomass supply on single biogas producers, farmers and farmland (B1, B2, B3). This would eventually feed back into social perceptions of the benefits of biogas and may reduce social opposition (B5).

The study shows that organizational capacity is quite crucial to improve the efficiency of the system (B7). Initiatives in this direction could be beneficial and at the same time reduce financially-risky investments by directly connecting different actors (e.g. local technicians). Cooperation might also prevent natural resources exploitation (e.g. land) but municipalities should intervene more proactively in land planning.

These policy initiatives would address some of the key underlying problems of this system; namely lack of cooperation, one-way policy solutions and lucrative policy measures. However, it is essential to bear in mind that natural systems have limits to growth and therefore natural resources such as land and agriculture byproducts should not overshoot their capacity to regenerate., Further research could simulate the system, estimate these limits and more effectively test these hypothetical interventions.

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